

Smart City Bike Sharing Station Performance Analysis

Project Overview: This Power BI project analyzes smart city bike-sharing station data to evaluate station utilization, bike availability, operational efficiency, and city-wise performance. The dashboard transforms raw station data into actionable urban mobility insights.

Objectives:

- Analyze bike station availability across cities
- Monitor operational and empty stations
- Evaluate station utilization efficiency
- Compare banking-enabled and non-banking stations
- Build an interactive analytical dashboard using Power BI

Assignment Tasks Completed:

- ✓Imported and explored dataset in Power BI
- ✓Cleaned and transformed data using Power Query
- ✓Built star schema data model with fact and dimension tables
- ✓Created DAX measures and calculated columns
- ✓Designed interactive dashboard with KPIs, charts, slicers, and map visualizations
- ✓Summarized insights and recommendations

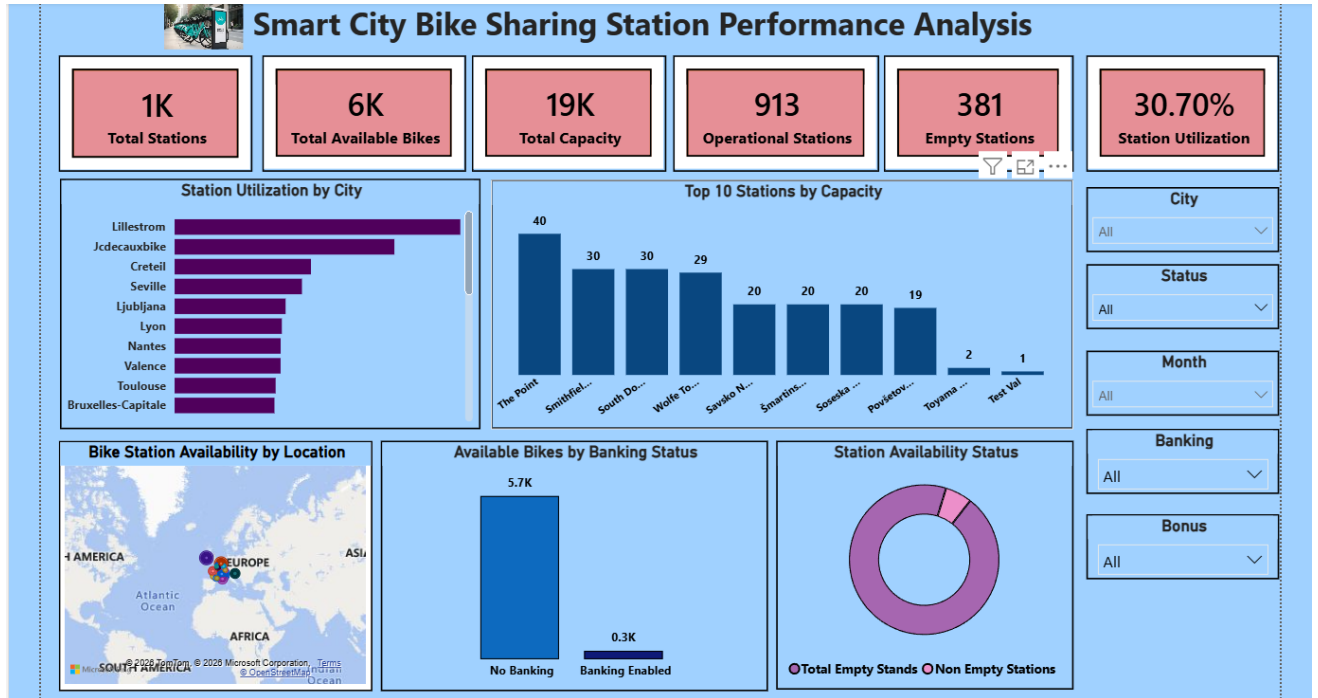
Key Insights:

- Total stations analyzed: 1K
- Total available bikes: 6K
- Overall station utilization: 30.70%
- Several stations remain underutilized despite high capacity
- Non-banking stations contain higher bike availability
- Empty stations indicate redistribution and balancing opportunities

Recommendations:

- Improve bike redistribution in underperforming stations
- Increase monitoring of empty stations
- Expand utilization strategies in low-performing cities
- Optimize operational planning using real-time monitoring dashboards

Dashboard Overview



Power Query Cleaning & Transformation

Table: Table.AddColumn("Inserted Year", "Month Name", each Date.MonthName([Date]), type text)

	Empty Stands	Available Bikes	Last Update	Date	Year	Month Name
1	1	0	0	02-06-2022 15:14:38	02-06-2022	2022 June
2	9	0	0	06-04-2022 12:54:35	06-04-2022	2022 April
3	15	0	0	27-09-2022 10:57:48	27-09-2022	2022 September
4	15	15	0	08-06-2022 11:50:13	08-06-2022	2022 June
5	0	0	0	23-05-2022 20:11:50	23-05-2022	2022 May
6	10	0	0	13-09-2022 11:28:01	13-09-2022	2022 September
7	24	24	0	06-09-2022 18:02:17	06-09-2022	2022 September
8	20	13	7	14-09-2022 13:27:09	14-09-2022	2022 September
9	10	0	0	12-09-2022 11:18:28	12-09-2022	2022 September
10	20	2	0	03-10-2022 11:34:01	03-10-2022	2022 October
11	10	10	0	13-09-2022 15:41:24	13-09-2022	2022 September
12	14	0	0	26-09-2022 16:44:57	26-09-2022	2022 September
13	15	0	0	03-10-2022 12:02:43	03-10-2022	2022 October
14	10	0	0	10-10-2022 12:40:37	10-10-2022	2022 October
15	8	0	0	26-09-2022 11:29:34	26-09-2022	2022 September
16	32	0	0	17-10-2022 11:36:42	17-10-2022	2022 October
17	16	0	0	24-10-2022 13:08:08	24-10-2022	2022 October
18	17	0	0	04-10-2022 11:49:03	04-10-2022	2022 October
19	1	0	1	30-08-2022 21:00:02	30-08-2022	2022 August
20	10	0	0	08-11-2022 14:53:04	08-11-2022	2022 November
21	14	0	0	22-11-2022 12:47:11	22-11-2022	2022 November
22						

Query Settings

PROPERTIES

Name: Sheet1

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Renamed Columns
- Changed Type1
- Split Column by Delimiter
- Changed Type2
- Renamed Columns1
- Removed Duplicates
- Replaced Value
- Trimmed Text
- Cleaned Text
- Capitalized Each Word
- Filtered Rows
- Uppercased Text
- Changed Type3
- Inserted Date
- Inserted Year
- Inserted Month Name

Model View / Star Schema

